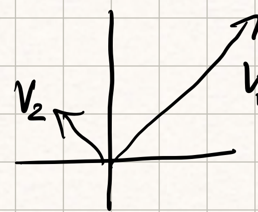
Back to the same linear function from last class:

$L(v) = \lambda v \qquad v_1 = (1,1) \ \lambda_1 = 3 \ \ v_2 = (-1,1) \ \lambda_2 = 1$



1) $[v_1, v_2]$ forms a basis

2) $\operatorname{Det} L = \lambda_1 \lambda_2$

how do we compute eigen things for arbitrary linear funcs

$$Lv = \lambda v = \lambda I v \implies Lv - \lambda I v = \vec{0}$$

$$(L - \lambda I)v = \vec{0}$$

$L - \lambda I$ is not invertible $\implies |L - \lambda I| = 0$

What are the eigenvalues of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = L$

$\begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0 \implies (1-\lambda)^2 = 0 \quad \lambda = 1$ So $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has 1 eigenvalue $\lambda = 1$

$\begin{bmatrix} 1-1 & 0 \\ 0 & 1-1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \vec{0}$ every vector in $\mathbb{R}^2$ is an eigenvector of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ w/ an eigenvalue of 1

__Ex__ find the eigenstuff of

$\begin{bmatrix} -2 & 1 \\ -2 & -1 \end{bmatrix}$ $\quad \begin{vmatrix} -2-\lambda & 1 \\ -2 & -1-\lambda \end{vmatrix} = 0 = (-2-\lambda)(-1-\lambda) + 2$

$\mathbb{R}^2 \to \mathbb{R}^2 \qquad = \lambda^2 + 3\lambda + 4 \quad \lambda = \frac{-3 \pm \sqrt{9 - 4(4)}}{2}$

$$\lambda = \frac{-3 \pm \sqrt{-7}}{2} = \frac{-3 \pm i\sqrt{7}}{2}$$

$\begin{bmatrix} -2 & 1 \\ -2 & -1 \end{bmatrix}$ has 2 complex eigenvalues

$$\underbrace{\begin{bmatrix} -2-\left(\frac{-3\pm i\sqrt{7}}{2}\right) & 1 \\ -2 & -1-\left(\frac{-3\pm i\sqrt{7}}{2}\right) \end{bmatrix}}_{} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\to \mathbb{C}^2 \to \mathbb{C}^2$

1) interperit this as the original $\mathbb{R}^2 \to \mathbb{R}^2$ question
2 eigenvalues in $\mathbb{C}$ and no eigenvectors

2) if we think of $\begin{bmatrix} -2 & 1 \\ -2 & -1 \end{bmatrix}$ as $\mathbb{C}^2 \to \mathbb{C}^2$ then there are 2 eigenvectors

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$ax^2+bx+c=0$ $a,b,c \in \mathbb{R}$ x is not guaranteed to be in $\mathbb{R}$
"not algebraically closed"

<u>Def</u>: A field F is algebraicaly closed if every polynomial equation w/ coefficients in F has solutions

$\mathbb{C}$ <u>is</u> algebraically closed

A map $\mathbb{R}^n \to \mathbb{R}^n$ may not have eigenvalues in $\mathbb{R}$
A map $\mathbb{C}^n \to \mathbb{C}^n$ always has eigenvalues